

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Hard

Question

The cost of renting a carpet cleaner is **\$52** for the first day and **\$26** for each additional day. Which of the following functions gives the cost $C(d)$, in dollars, of renting the carpet cleaner for d days, where d is a positive integer?

Answer

- A. $C(d) = 26d + 26$
- B. $C(d) = 26d + 52$
- C. $C(d) = 52d - 26$
- D. $C(d) = 52d + 78$

Correct Answer: A

Rationale

Choice A is correct. It's given that the cost of renting a carpet cleaner is **\$52** for the first day and **\$26** for each additional day. Therefore, the cost $C(d)$, in dollars, of renting the carpet cleaner for d days is the sum of the cost for the first day, **\$52**, and the cost for the additional $d - 1$ days, **\$26(d - 1)**. It follows that $C(d) = 52 + 26(d - 1)$, which is equivalent to $C(d) = 52 + 26d - 26$, or $C(d) = 26d + 26$.

Choice B is incorrect. This function gives the cost of renting a carpet cleaner for d days if the cost is **\$78**, not **\$52**, for the first day and **\$26** for each additional day.

Choice C is incorrect. This function gives the cost of renting a carpet cleaner for d days if the cost is **\$26**, not **\$52**, for the first day and **\$52**, not **\$26**, for each additional day.

Choice D is incorrect. This function gives the cost of renting a carpet cleaner for d days if the cost is **\$130**, not **\$52**, for the first day and **\$52**, not **\$26**, for each additional day.